Lesson 28: Plane Curves and Parametric Equations

Using TI-84 to graph parametric equations:

press [MODE], highlight PAR in the fourth row
press [ENTER].
Press [Y=] to enter equations
press [GRAPH] to see graph.

press [WINDOW] to adjust window settings (Tmin, Tmax, Tstep)
press [2nd] [GRAPH] to see table of values.

Using online 2D Calculator to graph parametric equations:
Example $x = 4t - 5$ and $y = 3t - 1$

Input (4t - 5, 3t - 1) or Curve(4t - 5, 3t - 1, -10, 10)

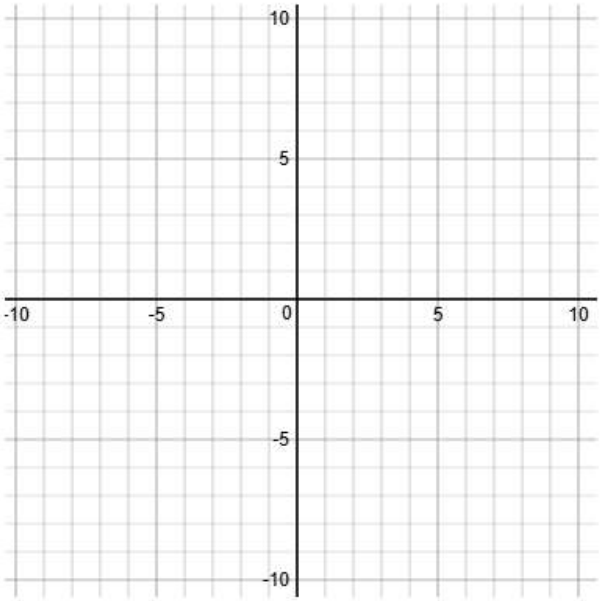
1) Parametric Equations: $x = 4t - 2, \quad y = 5t - 1$

a) Graph Parametric Equations.

b) Write the corresponding rectangular equation.
(multiple choice)

A) $y = 5\left(\frac{x-2}{2}\right) - 2$ B) $y = 5\left(\frac{x-3}{3}\right) - 1$

C) $y = 5\left(\frac{x-2}{4}\right) - 1$ D) $y = 7\left(\frac{x-2}{4}\right) - 1$



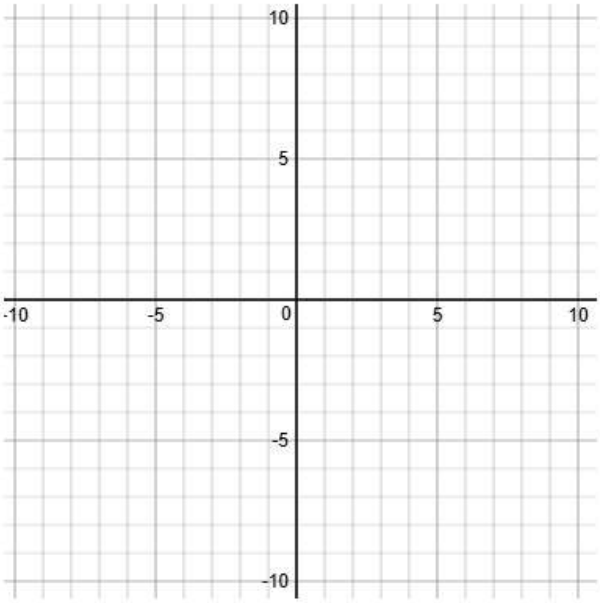
2) Parametric Equations: $x = 2t - 2, \quad y = t^2 + 1$

a) Graph Parametric Equations.

b) Write the corresponding rectangular equation. (multiple choice)

A) $y = \left(\frac{x-1}{2}\right)^2 + 1$ B) $y = \left(\frac{x-2}{2}\right)^2 + 1$

C) $y = \left(\frac{x-4}{2}\right)^2 + 1$ D) $y = \left(\frac{x-2}{2}\right)^2 - 5$



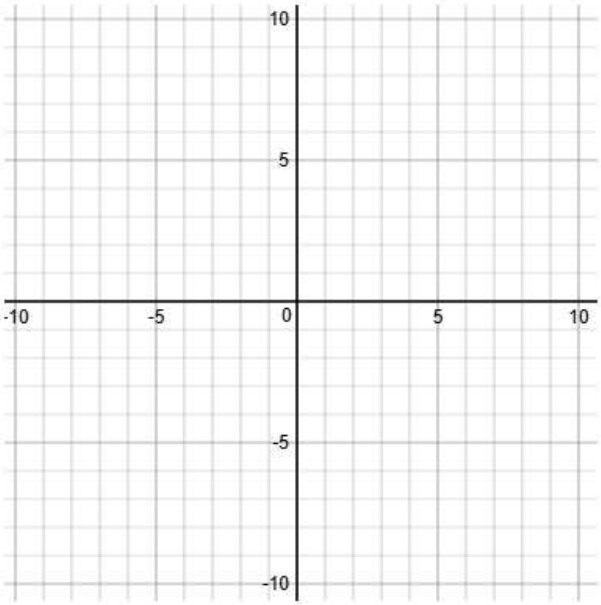
3) Parametric Equations: $x = 2t^2, \quad y = t^3 + 1$

a) Graph Parametric Equations.

b) Write the corresponding rectangular equation. (multiple choice)

A) $y = \left(\sqrt{\frac{x}{5}}\right)^3 + 1$ B) $y = \left(\sqrt{\frac{x}{4}}\right)^3 + 1$

C) $y = \left(\sqrt{\frac{x}{3}}\right)^3 + 1$ D) $y = \left(\sqrt{\frac{x}{2}}\right)^3 + 1$



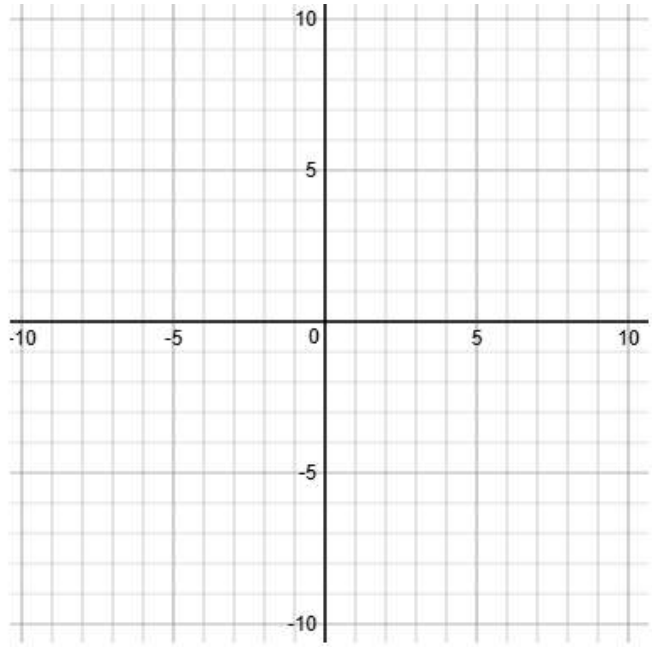
4) Parametric Equations: $x = \sqrt{t+4}, \quad y = t+1$

a) Graph Parametric Equations.

b) Write the corresponding rectangular equation. (multiple choice)

A) $y = x^2 - 4$ B) $y = x^2 - 3$

C) $y = x^2 - 2$ D) $y = x^2 - 1$



5) Parametric Equations: $x = t+4, \quad y = \dfrac{t}{t+1}$

Write the corresponding rectangular equation. (multiple choice)

A) $y = \dfrac{x-4}{x-3}$ B) $y = \dfrac{x-3}{x-3}$

C) $y = \dfrac{x-2}{x-3}$ D) $y = \dfrac{x-1}{x-3}$

6) Parametric Equations: $x = 3t+4, \quad y = |t+2|$

Write the corresponding rectangular equation. (multiple choice)

A) $y = \left|\dfrac{x-1}{3} + 2\right|$ B) $y = \left|\dfrac{x-2}{3} + 2\right|$

C) $y = \left|\dfrac{x-3}{3} + 2\right|$ D) $y = \left|\dfrac{x-4}{3} + 2\right|$

7) Parametric Equations: $x = e^t + 2, \quad y = e^{2t} + 5$

Write the corresponding rectangular equation. (multiple choice)

Hint: $\left(e^t\right)^2 = e^{2t}$

A) $y = (x-1)^2 + 5$ B) $y = (x-2)^2 + 5$

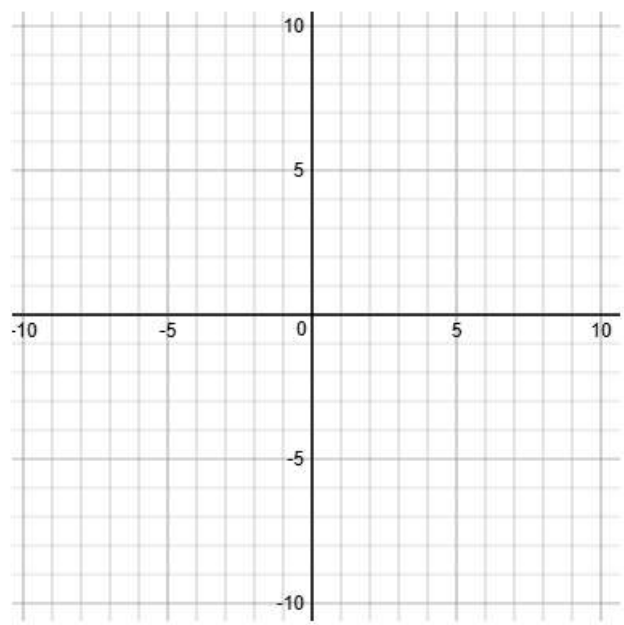
C) $y = (x-3)^2 + 5$ D) $y = (x-4)^2 + 5$

8) Parametric Equations: $x = 3\sec t, \quad y = 4\tan t$

a) Graph Parametric Equations.

b) Write the corresponding rectangular equation. (multiple choice)

Hint: $\sec^2 x - \tan^2 x = 1; \quad \csc^2 x - \cot^2 x = 1$



A) $\left(\frac{x}{3}\right)^2 + \left(\frac{y}{4}\right) = 1$ B) $\left(\frac{x}{3}\right)^2 - \left(\frac{y}{4}\right) = 1$

C) $\left(\frac{x}{4}\right)^2 - \left(\frac{y}{3}\right) = 1$ D) $\left(\frac{x}{2}\right)^2 + \left(\frac{y}{4}\right) = 1$

9) Parametric Equations: $x = t^2, \quad y = 4\ln t$

Write the corresponding rectangular equation. (multiple choice)

A) $y = 4\ln x$ B) $y = 4\ln x^2$

C) $y = 4\ln\sqrt{x}$ D) $y = 4\ln(4x)$

10) Parametric Equations: $x = 3\cos t, \quad y = 4\sin t$

a) Graph Parametric Equations.

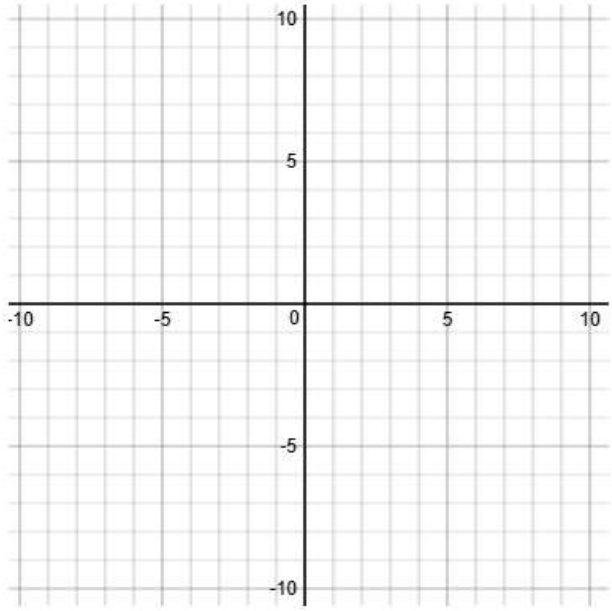
b) Write the corresponding rectangular equation.

(multiple choice)

Hint: $\sin^2 x + \cos^2 x = 1; \quad \sec^2 x - \tan^2 x = 1;$

A) $\left(\frac{x}{3}\right)^2 + \left(\frac{y}{4}\right) = 1$ B) $\left(\frac{x}{3}\right)^2 - \left(\frac{y}{4}\right) = 1$

C) $\left(\frac{x}{4}\right)^2 - \left(\frac{y}{3}\right) = 1$ D) $\left(\frac{x}{2}\right)^2 + \left(\frac{y}{4}\right) = 1$



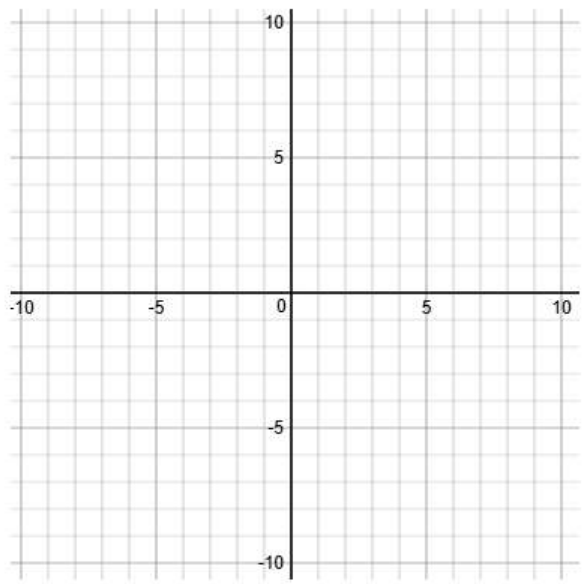
11) Parametric Equations: $x = -4 + 2\cos t, \quad y = 4 + 4\sin t$

a) Graph Parametric Equations.

b) Write the corresponding rectangular equation.

(multiple choice)

Hint: $\sin^2 x + \cos^2 x = 1; \quad \sec^2 x - \tan^2 x = 1;$

A) $\left(\frac{x+1}{2}\right)^2 + \left(\frac{y-1}{4}\right)^2 = 1$ B) $\left(\frac{x+2}{2}\right)^2 + \left(\frac{y-2}{4}\right)^2 = 1$

C) $\left(\frac{x+3}{2}\right)^2 + \left(\frac{y-3}{4}\right)^2 = 1$ D) $\left(\frac{x+4}{2}\right)^2 + \left(\frac{y-4}{4}\right)^2 = 1$